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Combining Domain Ontologies and BPMN Models at the CIM Level to generate IFML Models

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Abstract

The use of Business Process Model and Notation (BPMN) has been widely adopted by several research studies as one of the standard metamodels dedicated to the representation and design of business processes at the CIM level of MDA-oriented approaches. However, all of these researches don't include semantic data that accompanies business processes in the early stages of software development. Thus, the involvement of domain ontologies in this software development phase is key for a complete representation of the CIM level. This article presents a new modeling approach based on a model-driven development process, starting with the representation of BPMN and ODM models as CIM Models, then transforming them into an IFML diagram by applying a transformation engine. We can obtain easily a final code from the abstract model generated. A case study illustrates our approach. It represents a scenario of managing medications of an eHealth information system.

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Keywords: IFML, Ontology Modeling, CIM level, PIM, BPMN, Ontology Definition Metamodel (ODM), Model-driven Architecture (MDA).

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1. Introduction

The Model-driven Architecture (MDA) is an architecture based on models for the design of all complex software systems. This approach provides not only the ability to understand involved applications and the real world, but also a way to develop quickly and easily the applications through the definition of a development process going through three different layers [1–3]; the definition of Computation Independent Model (CIM), Then the Platform Independent Model (PIM), and finally the Platform Specific Model (PSM) which represents the closest model to the final code of the application. For the CIM level, several metamodels have been involved for its modeling, the most used is the BPMN standard [4]. As for the PIM level, UML diagrams are often used and recently a standard has been widely used, which is the IFML. IFML (Interaction Flow Modeling Language) is a user interfaces description language used to describe the content and interaction behavior of applications front-end [5,6]. For the PSMs, each approach uses a metamodel in accordance with the desired output platform.

The CIM describes the concepts of the business processes, terminology of the application domain and processes of the application, and hides the details related to its implementation. Most researchers focused on business models and ignored or missed the terminology in this level [7]. However, domain ontologies describe formally a structure of particular information and provide vocabularies. In addition, this concept has a similar level of the CIM layer of the model driven architecture. For this reason, we are interested in ontologies and their development due to the importance of integrating this concept in software systems modeling and their use in the early stages of their development. Our methodology is based on the representation and development of applications in a high level of abstraction, based on the two standards BPMN and ODM, thus allowing the implementation of a generation process of the PIM models representing consecutively IFML model in combination with UML domain model; that captures the content dependency. These output models will be then transformed into a code running on different platforms and devices. To validate our ideas and examine the applicability of our approach, we applied our approach to a case study of an Electronic Medical Records (EMR) application.

The structure of the remainder of this paper is as follows. In section 2, the related work will be discussed. Section 3 is devoted to the proposed approach and the MDD process, showing the experimental results on a case study. The conclusion is reported in section 4.

2. Related work

As it has been introduced, domain ontologies models have been widely studied and surveyed from different perspectives in model-driven engineering. as their use in PIM models [8–10], or their integration in PSM level [11,12] . Still, no research works propose their integration in the CIM level. Indeed, the CIM level accepts many metamodels, which has a well-formed behavioral and structural specification, they have been proposed by many research works. However, BPMN has used in the majority of model-based approaches. After analyzing these approaches, we have emphasized the following points: all these approaches start with a presentation of the business process, and in most cases, BPMN is used in this regard. as a second point, we noticed that the process of developing the CIM model to the PIM model consists of three steps. the first step is to define the CIM metamodel. In the second step, an M2M transformation is applied to produce a PIM model describing the CIM model analysis. Finally, the third step represents the generation of the PIM model conforming to the abstract metamodel. On the basis of these points, we have selected evaluation criteria to compare the different approaches analyzed. These criteria are as follows: the choice of the CIM metamodel, the transformation of the CIM models into PIM models and the choice of the PIM metamodel. Table 1 presents a comparative study, based on the characteristics mentioned above, of nine proposed approaches for modeling applications in the most abstract level of modeling independent of any implementation and technology.

These in-depth studies have allowed us to better understand the process of generating PIM models from CIM models. By analyzing the table 1, we notice that these approaches enable the presentation of CIM level, also the majority of them support the M2M Transformation, but there is no solution which permits to add a domain modeling view to business Models. Since no one has presented the ontologies in this level. However, unlike related works depicted in table 1, this work proposes a new approach that integrates the domain model in the CIM level and establishes a transformation of the CIM to PIM from the business and domain ontology models to IFML and UML class models.

Table 1. CIM to PIM approaches based on business models in the CIM level

The Studied approaches	Year of Publication	The CIM Models		CIM to PIM transformation		The PIM Models
		Business Model	Domain Model	Manual	Automatic generation	
Rhazali et al [13]	2018	Yes (UML Activity Diagram)	No	No	Yes (ATL)	UML Models
Arrhioui et al [14]	2018	Yes (BPMN)	No	No	Yes (QVT)	UML class diagram
Rhazali et al [15]	2016	Yes (BPMN)	No	No	Yes (ATL)	UML Diagrams (Use Case, State and Class diagrams)
Rhazali et al [16]	2016	Yes (BPMN)	No	No	Yes (ATL)	UML Models + SoaML & IFML Models
Kriouile et al [17]	2015	Yes (BPMN)	No	Yes	Yes (QVT)	UML Class model,
Rhazali et al [18]	2015	Yes (BPMN, UML Activity Diagram)	No	Yes	No	UML State diagram model
Sharifi et al [19]	2012	Yes (Business use case and analysis Models)	No	No	No	No
Kherraf et al [20]	2008	Yes (UML Activity Diagram)	No	Yes	No	UML Component and Class diagram
Rodríguez et al [21]	2007	Yes (BPMN)	No	No	Yes (QVT)	Use case diagrams

3. The proposed process

The process of the desired approach relies on OMG standards, we cite BPMN, ODM, IFML and UML standards. Indeed, to implement our approach, we have developed a model-based development process. As shown in figure 1, the process contains three stages. In the first stage, we start with the definition of the CIM metamodels and the PIM metamodels, then we construct a computation independant models according to CIM metamodels to establish the following stage. In the second stage, we launch the M2M transformation, so as to produce a model result describing the application in an abstract way. And we can eventually, go to the last stage by applying an M2T transformation that generates a code using generation engine developed in previous works.

We apply this process to the case study of e-health management systems: We design a domain ontology of an electronic medical record (EMR) web vocabulary for describing medications and patients. We associate it with the business process diagram. This ontology will be used to annotate the BPMN diagram that describes a medication or patient in the early stage of the development process of an e-health management application.

3.1. The CIM and PIM construction

Our proposal takes into account the static and dynamic business dimension in the CIM level, thanks to the use of both ODM and BPMN standards. These, will preserve the knowledge of the company during the transformation at the PIM level. In this work, we try to present a drug management process in an electronic health system. we first constructed the input model from the BPMN metamodel. The business process diagram is composed of Pool allowing the description of processes. The Process meta-class is composed of several graphical elements "GraphicalObject" having several subclasses describing the messages the flow of objects or connection events and so on. The figure 2 shows the business process model designed to model the chosen scenario representing drug management.

Concerning the second standard used to represent domain ontologies as a CIM Model, we employ the two metamodels of ODM, which are: OWL, that includes all concepts of RDFS and RDF, and RDFS for designing domain ontology. For constructing ODM components, it is necessary to affect a URI to each one for identification. We reference these URIs in the components via URIReference which has a "LocalName" and the referenced URI.

The proposed approach involves the IFML standard for representing the PIM level. This metamodel has been widely used by several works, we can cite [5,22–27]. The model instance of IFML consists of two sub models; the Interaction Flow Model and Domain Model. So we can represent the PIM Level with the two standards: IFML and UML

3.2. CIM to PIM transformation

We were based on operational transformation language (QVT) [28] defined by the OMG to establish our model to model transformation. For mapping ODM to UML we consider all the UML metamodel (class diagram) components to be represented in the ODM standard. The source model in our mapping is an ehealth Model respecting the OWL2 syntax. For the second CIM model, the table 2 defines the transformation rules from BPMN model to an IFML diagram.

Table 2. BPMN to IFML Transformation rules

Mapping rules	**Details**
SwimLane **to** ViewContainer	mapping smiwlane2viewContainer(sl :Swimlane, vc: ViewContainer): begin vc.name <- sl.label if sl.isType(XOR) vc.isXOR <- true end if end
Activity **to** ViewElement	mapping activity2ViewElement(ac :Activity, vc: ViewComponenet): begin vc.name <- ac.label If ac.dataInputAssociations.size() > 0 map ac.vc.viewComponentParts(ac.dataInputAssociations) else if ac.dataOutputAssociations.size() > 0 map ac.vc.viewComponentParts(ac.dataOutputAssociations) end if end
DataInputAssociaton **to** ViewComponentPart	mapping dataInputAssociation2VienComponentPart(dia:DataInputAssociation, vc: ViewComponenetPart) if dataInputAssociation.dataItems.size>0 foreach dataInputItem : dataInputAssociation.dataItems Object SimpleField{ this.name <- dataInputItem.label} end foreach end if end

DataOutAssociation **to** ViewComponentPart	mapping dataOutAssociation2VienComponentPart(doa:DataOutputAssociation, vc: ViewComponenetPart)
	if dataOutputAssociation.dataItems.size>0
	foreach dataOutputItem : dataOutputAssociation.dataItems
	Object VisualisationAttribute{
	this.name <- dataOutputtem.label
	this.value <- dataOutputItem.value
	}
	end foreach
	end if
	end

3.3. PIM generation

Once the medication management process has been sufficiently modeled using the BPMN and ODM metamodels, the CIM models are used as an input for the transformation engine developed for the approach. Indeed, we generate an IFML diagram as a PIM Model for the e-health management application that respects the IFML and UML metamodels. Our previous works, were proposed to generate easily and automatically the ontology file respecting the RDF/XML format **[11,12]**. The resulting owl file representing an e-health ontology studied in this article, helps to increase everyday productivity in big data. In that case, this ontology can be stored and implemented in many big data technologies such as Hadoop distributed file system **[11]**. Furthermore, This ontology plays a central role in the development of semantic web applications through the integration of annotations into web pages **[12]**.

4. Conclusion

We have proposed a new methodology for integrating ontologies at the most abstract level of MDA using the ODM standard. This work highlights the use of domain ontology as a CIM model of the MDA architecture, which is combined with business process models. Future works aim at integrating domain ontologies with other models in this level such as requirements models.

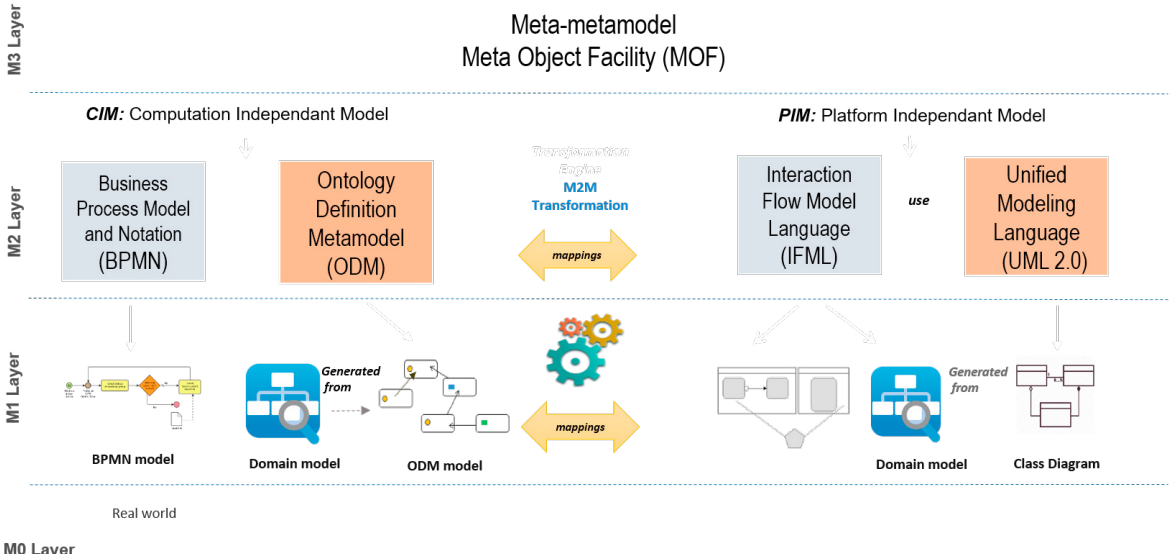


Fig. 1. Our proposal

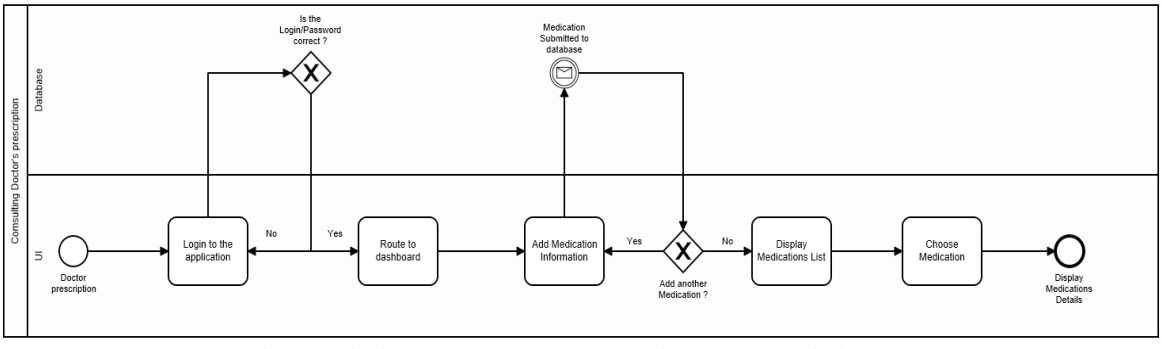


Fig. 2. Medication Management Process respecting BPMN Graphical Syntax

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